

Zecheng Gan

Advanced Materials Thrust of Function Hub

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Academic Appointments

Assistant professor, Thrust of Advanced Materials, HKUST(GZ), 2021–present

Assistant professor of mathematics (joint appointment), HKUST, 2022–present

Assistant professor (affiliate), department of mathematics, HKUST, 2021–2022

Postdoc, Courant Institute of Mathematical Sciences, NYU, 2019–2021

Advisor: Prof. Aleksandar Donev

Postdoc assistant professor, University of Michigan, Ann Arbor, 2016–2019

Advisors: Prof. Robert Krasny and Prof. Weihua Geng

Education

Ph.D., Applied & Computational Mathematics, SJTU, 2010-2016

Advisor: Prof. Zhenli Xu

B.S. Mechanical Engineering, SJTU, 2006-2010

Research Interests

Scientific computing and applied mathematics for problems arising from soft matter physics, fluid mechanics, and materials science. Molecular/Brownian dynamics, Monte Carlo methods, boundary integral equation methods, machine learning and data-driven methods.

Research Grants

PI, Natural Science Foundation of China, 01/01/2023–12/31/2025

NSFC Young Scientist fund (12201146)

PI, Natural Science Foundation of Guangdong province, 01/01/2023-12/31/2025

General fund (2023A1515012197)

PI, Guangzhou Municipal Science and Technology Bureau, 04/01/2023-03/31/2025.
Maiden Voyage Project (2023A04J0054)

PI, Guangzhou Municipal Science and Technology Bureau, 01/01/2024-12/31/2025.
Guangzhou-HKUST(GZ) Joint research project (2024A03J0606)

Educational Grants

PI, Guangzhou Education Bureau, 01/01/2026-01/01/2027.
Collaborative Education Program: Advanced Materials Innovative Talent Cultivation Plan.

Industrial Grants

Co-PI (with Prof. Tong-Yi Zhang), Industrial grants, Pipe China, 03/04/2025-05/31/2025.
Development of a Data-based Screening and Analysis Model for Circumferential Weld Cracks and Defective Welds in Long-Distance Pipelines.

PI (with Prof. Tong-Yi Zhang), Industrial grants, Pipe China, 01/01/2023-11/30/2023.
Small data prediction based on Gaussian process regression with improved accuracy and uncertainty estimation.

Internal Grants

PI (with UG student Chencheng Long), HKUST(GZ) Explorers Undergraduate Research Program (X Program), 08/01/2026-07/31/2027. *Numerical Study of Chemical-Field-Mediated Dynamics in Chiral Active Particles.*

PI (with UG student Xingyang Lin), HKUST(GZ) Explorers Undergraduate Research Program (X Program), 08/01/2026-07/31/2027. *Modeling and Simulation of Collective Behavior in Dielectrically Confined Charged Systems.*

PI (with UG student Xinjie Shi), HKUST(GZ) Explorers Undergraduate Research Program (X Program), 06/01/2025-05/31/2026. *Efficient kernel approximation: algorithms and scientific applications.*

Publications

Work in Preparing

1. with Leslie Greengard, Ning Wu, and Aleksandar Donev, Electrohydrodynamics of active colloids driven by AC electric fields.
2. with Sachin Natesh, Ondrej Maxian and Aleksandar Donev, Fast spectral methods for colloidal suspensions in doubly-periodic geometries.

3. with Weihua Geng and Robert Krasny, Computation of the Maxwell stress tensor and the electrostatic force between dielectric objects using boundary integrals.
4. with Zhiguo Yang, Shidong Jiang and Zhenli Xu, Fast algorithm for plasmonic interaction between nanospheres.

Preprints

5. with Qian Serena Hou, LSR-Net: Long-Short-Range Operator Learning for Pattern Dynamics on Manifolds. *Under review*.
6. with Z. Yang and R. Zhang, Spatial Disorder Can Enhance Temporal Order in a Hierarchical Active Matter. *Preprint*.
7. with Z. Yang, R. Zhang, et. al., A Fast Mesh-Free Method for Dynamics Simulations of Auto-chemotactic Droplets. *Preprint*.
8. with T. Hu, A spectral-subspace-augmented POD-Galerkin method for parametrized PDEs with limited snapshot data. *Under review*.
9. Yanyu Duan, Zecheng Gan, Ho-Kei Chan* and René Messina*, Dipolar cohesion in nascent helical packings. *Under review*.
10. with Qian Serena Hou, Inferring pattern formation from early-stage snapshots via Long-Short-Range neural network. *Under review*.
11. with Y. Duan, Designing Coulombic Contact Interactions between Polarizable Particles through Asymmetry. *Under review*.
12. with H. Song, et al., Improving Neural Operator by Kernel Decomposition in $d + 1$ Dimensions. *Under review*.
13. Ziwei Wang, Zecheng Gan, Shidong Jiang, Zhenli Xu and Erik Luijten, Anisotropic self-assembly of polarizable colloidal mixtures. *Preprint available upon request*.
14. Zecheng Gan, Weihua Geng and Robert Krasny, A boundary element method with extrapolation and singularity subtraction for charged dielectric spheres. *Preprint available upon request*.

Peer-Reviewed Journal Articles

(α - β indicates alphabetical author order, * indicates corresponding author, & indicates equal contribution.)

15. (α - β) **Zecheng Gan**, Xuanzhao Gao and Yuqing Li, An $\mathcal{O}(N)$ quasi-Ewald splitting method for nanoconfined electrostatics. *Accepted at SIAM Multiscale Modeling & Simulation, 2026*

16. Yanyu Duan, **Zecheng Gan**, Hervé Mohrbach, Ho-Kei Chan* and René Messina*, Dipolar cohesion in densely packed confined columns. *J. Chem. Phys.* 164(6), 064905, 2026.
17. Xuanzhao Gao[&], Qi Zhou[&], **Zecheng Gan*** and Jiuyang Liang*, Accurate error estimates and optimal parameter selection in Ewald summation for dielectrically confined Coulomb systems. *J. Chem. Theory Comput.*, 21(12), pp. 5890–5904, 2025.
18. (α - β) **Zecheng Gan**, Xuanzhao Gao, Jiuyang Liang and Zhenli Xu, Random batch Ewald method for dielectrically confined Coulomb systems. *SIAM J. Sci. Comput.*, 47(4), B846-B874, 2025.
19. Yanyu Duan and **Zecheng Gan***, Quantitative theory for critical conditions of like-charge attraction between polarizable spheres. *J. Chem. Theory Comput.*, 21(6), pp. 2822-2828, 2025.
20. Yanyu Duan, **Zecheng Gan*** and Ho-Kei Chan*, Mechanisms of electrostatic interactions between two charged dielectric spheres inside a polarizable medium: An effective-dipole analysis. *Soft Matter*, 21, 1860-1872, 2025.
Selected as Journal Cover (Featured article).
21. (α - β) **Zecheng Gan**, Xuanzhao Gao, Jiuyang Liang and Zhenli Xu, Fast algorithm for quasi-2D Coulomb systems. *J. Comput. Phys.* 524, 113733, 2025.
22. Y. Qiao, X. Chen, Z. Wang, W. Xu, **Z. Gan** and J. Wang*, Microwave-assisted catalytic upcycling of polyolefin wastes into hydrogen and carbon nanotubes over Pt-promoted Fe/Ni bimetallic catalysts. *Appl. Catal. B*, 384,126214, 2025.
23. Y. Xie, X. Lu, P. Bao, Y. Li, **Z. Gan**, D. Xiao, T.Y. Zhang* and J. Huang*, Estimating Heat Generation Rate of Cylindrical Lithium-Ion Batteries by a Transfer Function. *J. Electrochem. Soc.*, 172(8), pp. 080505, 2025.
24. X. Y. Chen, Z. Huang, **Z. Gan**, B. Dou, Y. Wang, X. Zhang and C. Yan*, Passivating deep-Level defects with Ag alloying to enhance the efficiency of Cu(In,Ga)(S,Se)₂ solar cells. *J. Phys. Chem. C*, 129(18), pp. 8537-8544, 2025.
25. Xuanzhao Gao and **Zecheng Gan***, Broken symmetries in quasi-2D charged systems via negative dielectric confinement. *J. Chem. Phys.* 161(1), 011102, 2024
26. Chung Wing Chan, Zheng Yang, **Zecheng Gan** and Rui Zhang*, Interplay of chemotactic force, Peclet number, and dimensionality dictates the dynamics of auto-chemotactic chiral active droplets. *J. Chem. Phys.* 161(1), 014904, 2024
27. Aref Hashemi*, Raul P Pelaez, Sachin Natesh, Brennan Sprinkle, Ondrej Maxian, **Zecheng Gan** and Aleksandar Donev, Computing hydrodynamic interactions in confined doubly periodic geometries in linear time. *J. Chem. Phys.* 158(15), 154101, 2023.

28. Jing Fu and **Zecheng Gan***, Harmonic surface mapping algorithm for electrostatic potentials in an atomistic/continuum hybrid model for electrolyte solutions. *Commun. Comput. Phys.*, 2021.
29. **Zecheng Gan**, Ziwei Wang, Shidong Jiang, Zhenli Xu and Erik Luijten*, Efficient dynamic simulations for charged dielectric spheres through a novel hybrid method. *J. Chem. Phys.* 151(2), 024112, 2019 **Selected for the 2019 JCP Editors' Choice.**
30. Huangfeng Ye, **Zecheng Gan**, Jae Wan Shim, Bo Kuang and Yanhua Yang*, On-node lattices construction using partial Gauss-Hermite quadrature for the lattice Boltzmann method. *Chin. Phys. B* 28(5), 054702, 2019
31. Manman Ma, **Zecheng Gan** and Zhenli Xu*, Ion structure near core-shell dielectric nanoparticle. *Phys. Rev. Lett.* 118(7), 076102, 2017
32. Xiaofei Guan, Manman Ma, **Zecheng Gan**, Zhenli Xu and Bo Li*, Hybrid Monte Carlo and continuum modeling of electrolyte with concentration-Induced dielectric variations. *Phys. Rev. E* 94(5), 053312, 2016
33. **Zecheng Gan**, Shidong Jiang, Erik Luijten and Zhenli Xu*, A hybrid method for systems of closely spaced dielectric spheres and ions. *SIAM J. Sci. Comput.* 38(3), B375-B395, 2016
34. **Zecheng Gan**, Huanxin Wu, Kipton Barros, Zhenli Xu and Erik Luijten*, Comparison of efficient techniques for the simulation of dielectric objects in electrolytes. *J. Comput. Phys.* 291, 317-333, 2015
35. **Zecheng Gan** and Zhenli Xu*, Efficient implementation of the Barnes-Hut octree algorithm for Monte Carlo simulations of charged systems. *Sci. China Math.* 57(7), 1331-1340, 2014
36. **Zecheng Gan**, Xiangjun Xing and Zhenli Xu*, Effects of image charges, interfacial charge discreteness, and surface roughness on the Zeta potential of spherical electric double layers. *J. Chem. Phys.* 137(3), 034708, 2012
37. **Zecheng Gan** and Zhenli Xu*, Multiple-image treatment of induced charges in Monte Carlo simulations of electrolytes near a spherical dielectric interface. *Phys. Rev. E* 84(1), 016705, 2011

Organized Academic Activities

Organizer, AMAT Public Research Seminar Series. Speakers: Bowei Wu (Univ. of Massachusetts, Lowell, 2025.06.12); Qingtang Su (Chinese Academy of Sciences and Morningside center of Mathematics, 2025.06.12)

Co-organizer (with Prof. Weijia Wen (Chair) et al.), The 6th Greater Bay Area Soft Matter and Life Matter Forum. (Jan. 9-13, 2025).

Co-organizer (with Prof. Ping Gao (Chair), et. al.), Cross-Campus Workshop on Chemistry and Materials Research. [link](#).

Organizer, Public talk on “Successful publishing with ACS and Nano Letters”; Speaker: Elena Besley (Head of Materials Chemistry, University of Nottingham; and Associate editor at Nano Letters). [link](#).

Organizer (with Profs. Ho-Kei Chan), Mini-workshop on Electrostatics in Advanced Materials (2024); Speakers: Elena Besley (University of Nottingham), Rene Messina (University of Lorraine), Decai Huang (NJUST), etc. [link](#). **The workshop was reported on the newsletter of Institute of Physics (IOP).** [link](#).

Organizer (with Dr. Yuqing Li), Minisymposium on “Modeling and Simulation for Materials Science and Deep Learning”, EASIAM 2024. [link](#).

Co-organizer (with Prof. Ping Gao (Chair), et. al.), Cross-Campus Workshop on Biology, Energy and Materials. [link](#).

Organizer (with Profs. Ho-Kei Chan and Xiaoxiao Wu), Mini-workshop on Electrostatics in Advanced Materials; Speakers: Rene Messina (University of Lorraine), Erdal C. Oğuz (Institute of Physics, Chinese Academy of Sciences), etc. [link](#).

Organizer, AMAT Public Research Seminar Series on Computational Sciences; Speakers: Bowei Wu (Univ. of Massachusetts, Lowell, 2023.07.25); Qingtang Su (Chinese Academy of Sciences and Morningside center of Mathematics, 2023.07.28)

Organizer (with Profs. Shidong Jiang and Yejun Gu), Minisymposium on “Simulations and Algorithms for Materials Sciences”, 10th International Congress on Industrial and Applied Mathematics (ICIAM), 2023 Tokyo, [link](#).

JuliaCN 2022 Meet-up, 2022.12, with Jinguo Liu, et. al., [link](#).

Outreach Activities

Popular science lecture for international High School students, EDGE Program (Elite Discovery & Global Exploration), July. 22th, 2026.

Popular science lecture on “Materials science & AI” to Guangdong Experimental High School students, May. 14th, 2025.

Popular science lecture on “Materials science & AI” to Chaoyang experimental high school students (joint with Prof. Ping Gao), Jan. 21th, 2025.

Co-organizer (with Guofu Tang and Prof. Yi-lung Kuo), the “math source tour at UIC”, Oct. 22nd, 2024.

Open Source Softwares

Co-Developer/Maintainer for

StochasticHydroTools: a toolbox for stochastic electro-hydrodynamic problems.

TABI-P: treecode-accelerated boundary integral Poisson solver for biomolecules.

HybridMD: molecular dynamics package for colloidal spheres accelerated by hybrid method.

Conference and Workshop Presentations

CSIAM conference on big data & AI (CSIAM-BDAI 2026), Xiamen, July 17th-July 19th, 2026

International Conference on Scientific Computation and Differential Equations (SciCADE 2026), University of Edinburgh, June 29th-July 3rd, 2026

15th Asian Control Conference (ASCC), Bali, Indonesia, June 17-21, 2026

The 15th national conference on inverse problems, imaging and their applications, Wenzhou-Kean university, May. 16-17, 2026.

Workshop on Recent Advances in Quasi-Monte Carlo and Related Topics, SCUT, Dec. 4-5, 2025.

The 4-th International Symposium on Computational Energy Science (COMPES2025), TJU, Oct. 10-13, 2025.

International workshop on “Electrostatics in Soft Matter”, NJUST, Sep. 22, 2025.

Workshop on “Calibrating prediction uncertainty : statistics and machine learning perspectives”, Isaac Newton Institute for Mathematical Sciences, Cambridge University, June 1-7, 2025.

2025 DKU Soft Matter Symposium - From “Microscopic” interactions to Collective Behaviors, Duke Kunshan University, May 17-19, 2025.

Workshop on mathematics for science and engineering, SJTU, Dec.14-15, 2024.

Workshop on applied and computational mathematics, Guangdong University of Technology, Dec.12-13, 2024.

Workshop on scientific computing and machine learning, SJTU, Dec.6-8, 2024.

Cross-Campus Workshop on Chemistry and Materials Research, HKUST(GZ), Nov.16, 2024.

CSIAM annual conference, Nanjing, Oct. 25-17, 2024.

Conference of Scientific Machine Learning (CSML2024), Shanghai, Aug. 9-12, 2024.

2024 Joint Annual Conference of Physical Societies in Guangdong-Hong Kong-Macao Greater Bay Area, Macao, Jul 30 - Aug 2, 2024

International Conference on Scientific Computation and Differential Equations (SciCADE 2024), National University of Singapore, July 15th-19th, 2024

East Asia Section of SIAM (EASIAM 2024), University of Macau, June 28th-July 1st, 2024

Quantum Materials & Materials Informatics Symposium, HKUST(GZ), 2023

Workshop on Recent Advances in Fast algorithms, SICIAM, CUHK(SZ), 2023.8.2

Mathematics in Action (MiA2023): Modeling and Analysis in Biology and Material Sciences, SJTU, Duke Kunshan and Soochow Univ., 2023.6

Symposium on Applied Mathematics and Computational Mathematics, Sun Yat-Sen University, 2023.6

Optimization algorithms for data science, CSIAM annual meeting, 2022.11

Scientific computing workshop, CUHK(SZ) and Shenzhen research institute of big data, 2022.10

Workshop on Machine Learning and Data-Driven Scientific Computing, BNU-UIC, 2022.1

Numerical Methods for Incompressible Fluid Systems, SIAM Annual Meeting (virtually), 2021.7

Advances in numerical methods, theories and applications, CUHK(SZ), 2021.6

Engineering Sciences Youth Forum, College of Engineering, Peking University, 2021.2

Hausdoff school 2020–MCMC: Recent developments and new connections, Hausdoff center for mathematics, University of Bonn, Bonn, Germany, 2020.3.28-2020.4.3 (shifted online due to COVID-19)

2019 Applied Math Youth Forum, SMS & BICMR, Peking University, 2019.12

Protein Electrostatics workshop, TSRC, Telluride, Colorado, USA, 2019.6

Mathematical Fluids, Materials and Biology—A conference in honor of the 60th Birthday of Michael J. Shelley (poster), UMich, USA, 2019.6

SIAM Great Lakes Section Meeting, UMich, USA, 2019.4

Midwest Numerical Analysis Day 2019, IIT, USA, 2019.4

The 2019 MICDE Symposium - Computational Science: Classical Origins, New Frontiers (poster), UMich, USA, 2019.4.10

SIAM Conference on Computational Science and Engineering, Washington, USA, 2019.2

2018 Young Mathematician Forum, Sun Yat-sen University, December 25, 2018

AMS Fall Central Sectional Meeting, UMich, USA, 2018.10

SIAM Annual Conference 2018, Portland, USA, 2018.7

The 2018 MICDE Symposium - Computation: A Pillar of Science and a Lens to the Future (poster), UMich, USA, 2018.3

The 14th Annual Conference on Frontiers in Applied and Computational Mathematics, NJIT, USA, 2017.6

The 2017 MICDE Symposium - The New Era of Data-Enabled Computational Science (poster), UMich, USA, 2017.4

SIAM Conference on Computational Science and Engineering (poster), Atlanta, USA, 2017.2

The International Congress on Industrial and Applied Mathematics (poster), Beijing, China, 2015.8

The 1st INS-ZY Student Conference On Natural Sciences, Shanghai, China, 2015.4

10th Conference on Computational Sciences and Engineering, Shanghai, China, 2014.12

9th National Physics Conference on Soft Matter and Life Science, Wenzhou, China, 2014.11

Frontiers of Soft Matter Physics: from Non-equilibrium Dynamics to Active Matter (poster), Hong Kong, 2014.01

9th Conference on Computational Sciences and Engineering, ECUST, Shanghai, China, 2013.11

Workshop on Computational Challenges in Complex Charged Systems, Shanghai, China, 2013.09

Workshop on Monte Carlo Methods in the Physical and Biological Sciences (poster), ICERM, Brown University, USA, 2012.10

Workshop and Summer School on Coulomb Many-body Systems (poster), Shanghai, China, 2012.06

Seminar Talks

Research seminar, HIT(SZ), 2023.7.27

Materials informatics seminar series, Guangzhou key lab of materials informatics, 2022.10

Applied Mathematics Seminar, Shenzhen International Center for Industrial and Applied Mathematics, 2021.11

Seminar series, Department of Physics, HIT(SZ), 2021.11

Seminar series in mathematical sciences, Institute of Mathematical Sciences, ShanghaiTech, 2021.6

Modeling and simulation methods in softer matter physics, SJTU, 2021.6

MRSEC biweekly seminar, NYU, 2021.2

Mathematical Research Seminar Series, Zu Chongzhi Center for Mathematics and Computational Sciences, Duke University (Kunshan), 2021.1

Computational and Applied Mathematics (CAM) seminar, SMS, Peking University, 2020.12

Modeling and Simulation (MSG) seminar (with Ondrej Maxian and partner: Aleksandar Donev), Courant Institute, NYU, 2020.10

Modeling and Simulation (MSG) seminar (partner: Chris Miles), Courant Institute, NYU, 2019.10

Mathematics Symposium (Graduate Recruitment Weekend), UMich, 2019.3

Applied & Interdisciplinary Mathematics (AIM) seminar (substituted by Dr. Robert Krasny), UMich, 2019.3

Fast Algorithms for the Electrostatics Between Charged Dielectric Objects, BNU, 2018.12

BIE methods for the Electrostatics in Biomolecule Simulations, INS, SJTU, 2017.7

BIM for the Electrostatics in Biomolecule Simulations, math department, TJU, 2017.7

New Faculty Capsule Research Talk, Department of Mathematics, UMich, 2016.8

Dielectric Effect in Charged Soft Matter Systems, NVIDIA Webinar Series, NVIDIA, 2016.6

Hybrid Method for Solving Electrostatics in Systems of Densely Packed Dielectric Spheres and Ions, Department of Mathematics, UMich, 2015.12

Self-assembly Structure of Binary Dielectric Colloidal NPs, CSML, Northwestern University, 2015.11

Fast Algorithms for Solving the Poisson's Equation with Close Compacted Dielectric Spheres and Ions, Computational Soft Matter Lab, Northwestern University, 2015.10

9th Doctoral Forum of Shanghai Jiao Tong University (Best Report Award), 2012.05

Topical Review on Enhanced Sampling Monte Carlo algorithms, SJTU, 2012.02-2012.05

Research Advising and Mentoring

Postdoc Advising

2026-now, Yue Fang (PhD from HKUST, clear water bay), co-supervising with Prof. Jiaying Yuan, HKUST (AMAT, GZ)

Doctoral Advising

2025-now, Shiyi Bai (previous RBM student), 1st supervisor, co-supervising with Prof. Jiaying Yuan, HKUST (AMAT, GZ)

2024-now, Kaiwen Jin, 1st supervisor, co-supervising with Prof. Jin-Guo Liu, HKUST (AMAT, GZ)

2024-now, Xuanyan Chen (previous RBM student), 1st supervisor, co-supervising with Prof. Chang Yan, HKUST (AMAT, GZ)

2024-now, Qian Hou, 1st supervisor, co-supervising with Prof. Jun Wang, HKUST (AMAT, GZ)

2023-now, Pengxiao Bao, 2nd supervisor, co-supervising with Prof. Tong-yi Zhang, HKUST (AMAT, GZ)

2023-now, Songyuan Geng, 2nd supervisor, co-supervising with Prof. Haoxiang Li, HKUST (AMAT, GZ)

2023-now, Weitao Lan, 1st supervisor, co-supervising with Prof. Ruiting Zuo, HKUST (AMAT, GZ)

2023-now, Guifa Xu, 2nd supervisor, co-supervising with Prof. Jun Wang, HKUST (AMAT, GZ)

2023-now, Boran Zhao, 2nd supervisor, co-supervising with Prof. Xiakun Chu, HKUST (AMAT, GZ)

2023-now, Zichun Wang, 2nd supervisor, co-supervising with Prof. Ruiting Zuo, HKUST (Fintech, GZ)

2022-now, Yanyu Duan, 1st supervisor, co-supervising with Prof. Xiaoxiao Wu, HKUST (AMAT, GZ)

2022-now, Tianhao Hu, 1st supervisor, co-supervising with Prof. Yang Xiang, HKUST (AMAT, GZ)

2022-now, Yongmin Luo, 2nd supervisor, co-supervising with Prof. Jiaying Wu, HKUST (AMAT, GZ)

2022-now, Chenzhen Xu, 2nd supervisor, co-supervising with Prof. Xiakun Chu, HKUST (AMAT, GZ)

2022-now, Zhigao Xie, 2nd supervisor, co-supervising with Prof. Chee-keong Tan, HKUST (AMAT, GZ)

2022-now, Yan Wang, 2nd supervisor, co-supervising with Prof. Chee-keong Tan, HKUST (AMAT, GZ)

2021-now, Zheng Yang (1st supervisor, co-supervising with Rui Zhang), HKUST (IIP, CWB)

2021-2025, Xuanzhao Gao (1st supervisor, co-supervising with Profs. Jin-Guo Liu and Yang Xiang), HKUST (IIP, CWB)

2020-2024, Jiayi Wang (2nd supervisor, co-supervising with Ping Gao), HKUST (IIP, CWB)

M. Phil. Advising

2024-now, Lingchuan Tan (co-supervise, with Prof. Huachen Cui), HKUST (GZ)

2023-now, Shiyi Bai, (main-supervise, with Prof. Jiaqiang Huang), HKUST (GZ) HKUST (GZ)

2022 summer intern, now RBM student, Wenjun Li, HKUST (GZ)

2021-2023, Wing Chun (Kyle) CHAN (co-supervise, with Prof. Rui Zhang), HKUST (Physics)

Undergraduate Advising

2025-now, RC1, 8 students from 5A-4F-8/10, co-supervising with Prof. Yunlong Zi.

2024-now, Guofu Tang, Yan Yang, Wei Yang.

Teaching Experience

Hong Kong University of Science and Technology, Guangzhou

Co-Instructor, AMAT2010 (Transport Phenomena in MSE): Spring 2026 (Co-teaching with Prof. Ping GAO)

Co-Instructor, AMAT1010 (Academic Orientation in MSE): Spring 2026 (Co-teaching with Prof. Ping GAO and AMAT faculties)

Instructor, UFUG1105 (Honors Calculus I): Fall 2023, Fall 2024, Fall 2025, Fall 2026 (Member of the course develop team at HKUST(GZ). I am also the course-coordinator for UFUG1105 in Fall 2024.)

Co-Instructor, AMAT6000A (Advanced Materials Informatics): Spring 2025 (Co-teaching with Prof. Tong-Yi Zhang)

Co-Instructor, UCMP6010 (Research Method): Spring 2023/2024/2025, Fall 2023/2024 (Member of the course develop team at HKUST(GZ))

Instructor, AMAT5215 (Mathematical Models for Interdisciplinary Sciences): Spring 2024 (Course owner at HKUST(GZ).)

Instructor, AMAT5250 (Mathematical Methods for Material Science): Fall 2022, Fall 2024 (Course owner at HKUST(GZ))

Instructor, AMAT5200 (Machine Learning for Material Science): Spring 2022, Fall 2023 (Course owner at HKUST(GZ))

Co-Instructor, FUNH5200 (Introduction to Materials Informatics II): Fall 2023 (Co-teaching with Prof. Tong-Yi Zhang)

Co-Instructor, FUNH5000 (Introduction to Function Hub for Sustainable Future): Spring 2024 (Co-teaching with Profs. Yang Huang, CK Tan and Peng Chen)

Coordinator, FUNH6800 (Function Hub Graduate Student Seminar): Fall 2021

NYU Shanghai

Recitation Instructor for “go-local” students due to the COVID-19,
Math for Economics 1 and 2: 2020–2021

Recitation Instructor for “go-local” students due to the COVID-19,
Multivariable Calculus: 2021

University of Michigan

Instructor, Math 286 (Honors Differential Equations): Winter 2019, Fall 2017

Instructor, Math 156 (Applied Honors Calculus II): Fall 2018

Instructor, Math 285 (Honors Calculus III): Winter 2017

Instructor, Math 115 (Calculus I): Fall 2016

Tutor, Umich Math Lab: Fall 2016

Instructor, Umich Math Circle: Creative Combinatorics I & II, October 2016

Shanghai Jiao Tong University

TA, Computational Statistics: Spring 2012, Spring 2014

Co-instructor, Probability and Statistics: Spring 2013, Fall 2013

TA, Scientific Computing: Fall 2011

TA, Calculus: Spring 2011

TA, Linear Algebra: Fall 2010

Professional Service

University Administration/Service (HKUST)

Science & Technology Festival Day Exhibition, Guangdong Guangya High School, Guangzhou, 2026.05.15

Member of the AI-Academic Committee for the “Major + AI” extended major program, HKUST(GZ), 2026-now.

Undergraduate Admissions Open Day, HKUST(GZ), 2026.04.04

Undergraduate Admissions Open Day, HKUST(GZ), 2026.02.08

Member of the Review and Selection Committee for the *Climbing Program* (special fund for Guangdong Undergraduate and Postgraduate Innovation Training), HKUST(GZ), 2026.01

UG coordinator, Materials Science & Engineering major, HKUST(GZ), 2025.09 - now
Undergraduate Admissions Open Day, HKUST(GZ), 2025.11.29

Undergraduate Admissions Open Day, HKUST(GZ), 2025.10.2

Member of College Entrance Examination (Gao Kao) Admissions Committee (Shanghai Area), 06.21-07.01, 2025.

Sub-committee leader (with Prof. Ting Fang) of the Red Bird M.Phil. Selection & Interview Committee (University Level), HKUST(GZ), 2025-2026.

Member of the Red Bird M.Phil. Selection & Interview Committee (University Level), HKUST(GZ), 2021-2022, 2023-2024, 2024-2025.

Promotion event speaker for undergraduate program and MSE major, Yangjing High School, Jianping High School and No.2 High School Of East China Normal University, 04.15-04.18, 2025.

Promotion event speaker for Red Bird Mphil program and AMAT thrust, Fudan University, 2025.03.31.

Promotion event speaker for Red Bird Mphil program and AMAT thrust, Shanghai Jiao Tong University, 2025.03.30.

Guest lectures for FUNH5000 on “AI for AMAT” (with no teaching credit), 4 lectures (2 hours each), hosted by Prof. Qichun Yang, HKUST(GZ), Fall 2024, and Spring 2025.

Advisor for the undergraduate “Math source society”, HKUST(GZ), Sep. 2024 - now.

Session chair on “Computational and Mathematical Methods”, Cross-Campus Workshop on Chemistry and Materials Research, HKUST(GZ), 2024.11

Session chair on “Materials Informatics”, Cross-Campus Workshop on Bio, Ene, and Mat, HKUST(GZ), 2023.11

Member of the Review and Selection Committee for Natural Science Foundation of Guangdong province (general program), HKUST(GZ), 2024.07

Co-organizer and Chair for AI&Materials Informatics session at QMMI conference, HKUSTGZ, 2023.12. *Invited speakers: Yuqing Li (SJTU), Jiaxing Yuan (Tokyo Univ.)*

Organizer for AMAT seminar series on computational sciences, HKUSTGZ, 2023-now. *Invited speakers: Bowei Wu(UMass), Qingtang Su (Morningside center of mathematics, CAS), Tao Luo and Yuqing Li (SJTU), Dong Wang (CUHK-SZ)*

Member of the Review and Selection Committee for Natural Science Foundation of Guangdong province (general program), HKUST(GZ), 2023.06

Member of the Senate Committee on Teaching and Learning Quality, HKUST(GZ), 2023-

now.

Committee Member of the *Green e Materials* theme-based lab, HKUST(GZ), 2022-now.

Committee Member of the *Materials Informatics* Guangzhou city key lab, HKUST(GZ), 2022-now.

Member of the Curriculum Development and Review Committee, Advanced Materials Thrust, HKUST(GZ), 2022-now.

Member of the External Liaison and Promotion Committee, Advanced Materials Thrust, HKUST(GZ), 2022-now.

Promotion events speaker for HKUST(GZ): Guangzhou, Shenzhen, Nanjing (2 events) and Shanghai (3 events), 2021-2022.

Extramural Administration/Service

Chair/Co-organizer for student mini-workshop on Scientific computing: algorithm and modeling, SJTU, 2023.6

Chair for session “Computational Electromagnetics”, SIAM Conference on Computational Science and Engineering , Washington, USA, 2019.2

Editorial board

Youth Editorial Board, *Journal of Materials Informatics*, 2024–now

Journal refereeing

Journal referee for: *Mathematics of Computation*, *SIAM Journal on Scientific Computing*, *Journal of Computational Physics*, *Journal of Scientific Computing*, *Accounts of Chemical Research*, *Physical Review Letters*, *Nano Letters*, *The Journal of Physical Chemistry Letters*, *PRX Energy*, *Macromolecules*, *Physical Review A*, *Physical Review E*, *Journal of Statistical Physics*, *Journal of Physical Chemistry*, *Molecular Systems Design & Engineering*, *Advances in Applied Mathematics and Mechanics*, *Scientific Reports*, *Applied Mathematics Modelling*, *Transport Phenomena*, *Frontiers in Applied Mathematics and Statistics*.

Professional Societies

Member, China Society for Industrial and Applied Mathematics (CSIAM), 2021–Present.

Member, American Physical Society (APS), 2020–Present.

Member, Society for Industrial and Applied Mathematics (SIAM), 2017–Present.

Member, Institute of Physics (IOP), UK, 2026–Present.

Selected Honors and Awards

Early Career Physicist (Nominated by the Dielectrics and Electrostatics Group Committee), Institute of Physics, UK, 2026

Best presentation award, The 4-th International Symposium on Computational Energy Science, 2025

Distinguished guest lecturer, Guangdong Experimental High School, 2025

National Scholarship, Ministry of Education, China, 2014-2015

AMD High Performance Computing Scholarship, HPC Center of Shanghai Jiao Tong University, 2013-2014

Postgraduate Academic Excellence Scholarship, Shanghai Jiao Tong University, 2012-2013

The *Cyrus Tang* Foundation Scholarship, Shanghai Jiao Tong University, 2006-2010

Research Impact and Technology Transfer

Quasi-2D electrostatic error analysis adopted in LAMMPS PR #4939(2026), enabling automatic slab-size selection for Ewald/PPPM simulations.

Co-developer of AIPipe V1.0, an AI-based pipeline-data analysis and prediction platform; software copyright disclosure filed at HKUST(GZ) (2025).

Developer/maintainer of open-source scientific-computing packages, including StochasticHydroTools, TABI-P, HybridMD, ExTinyMD and SolarLabs.